

Markscheme

Mathematics: analysis and approaches
Practice question bank

95. (a)

Consider the statement: " $2^n > n^2$ for every integer n satisfying $5 \leq n \leq 8$."

Verify the statement is true by checking each case (proof by exhaustion), and hence state the smallest value of n in this range for which $2^n - n^2 > 100$.

check each case: $n = 5 : 32 > 25, n = 6 : 64 > 36, n = 7 : 128 > 49, n = 8 : 256 > 64$, so the statement holds (M1) A1

compute $2^n - n^2$ for each: 7, 28, 79, 192 (M1) A1

the first value exceeding 100 occurs at $n = 8$ A1

[5 marks]